ENF7010 기후변화의 거시경제학

[2 강]
장기성장에 관한 몇가지 사실

(1) 좋은 모형의 조건
(2) 몇가지 전형적인 사실들

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좋은 모형(이론)의 원리

1. 세상을 닮아야.

2. Occam’s razor



1. 세상을 최소 이 정도는 닮아야: 전형적인 사실 (stylized facts)

어떤 세상? 경제

어느 경제? 지구, 또는 미국

왜 미국?

2. 모형 내에서 설명되어야 할 최소 범위

- 외생적인 (exogenous)
  originating/determined from outside a system

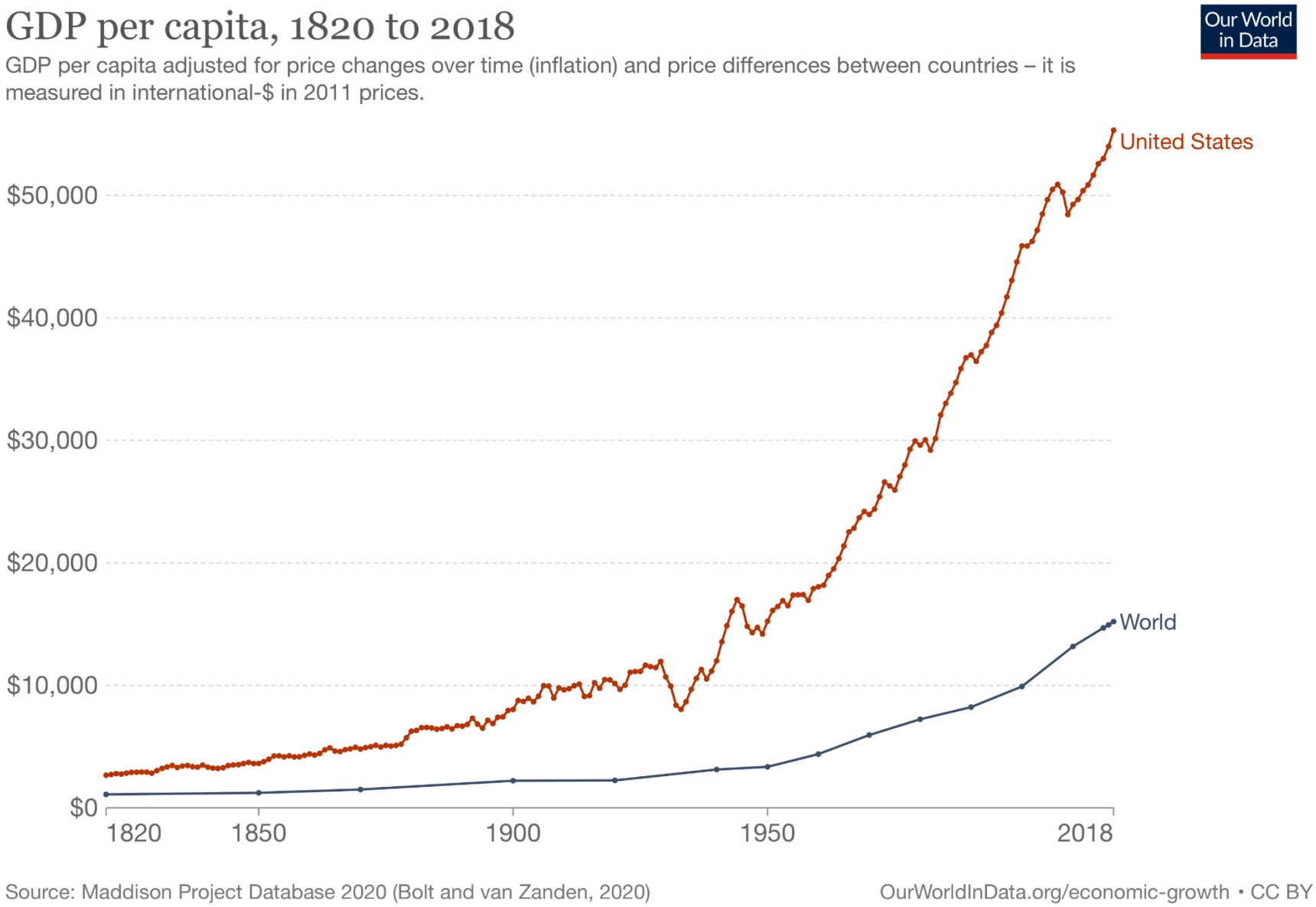
- 내생적인 (endogenous)
  originating/determined within a system

전형적인 사실들 (Stylized facts; Kaldor facts)

Hassler and Krusell (2018)

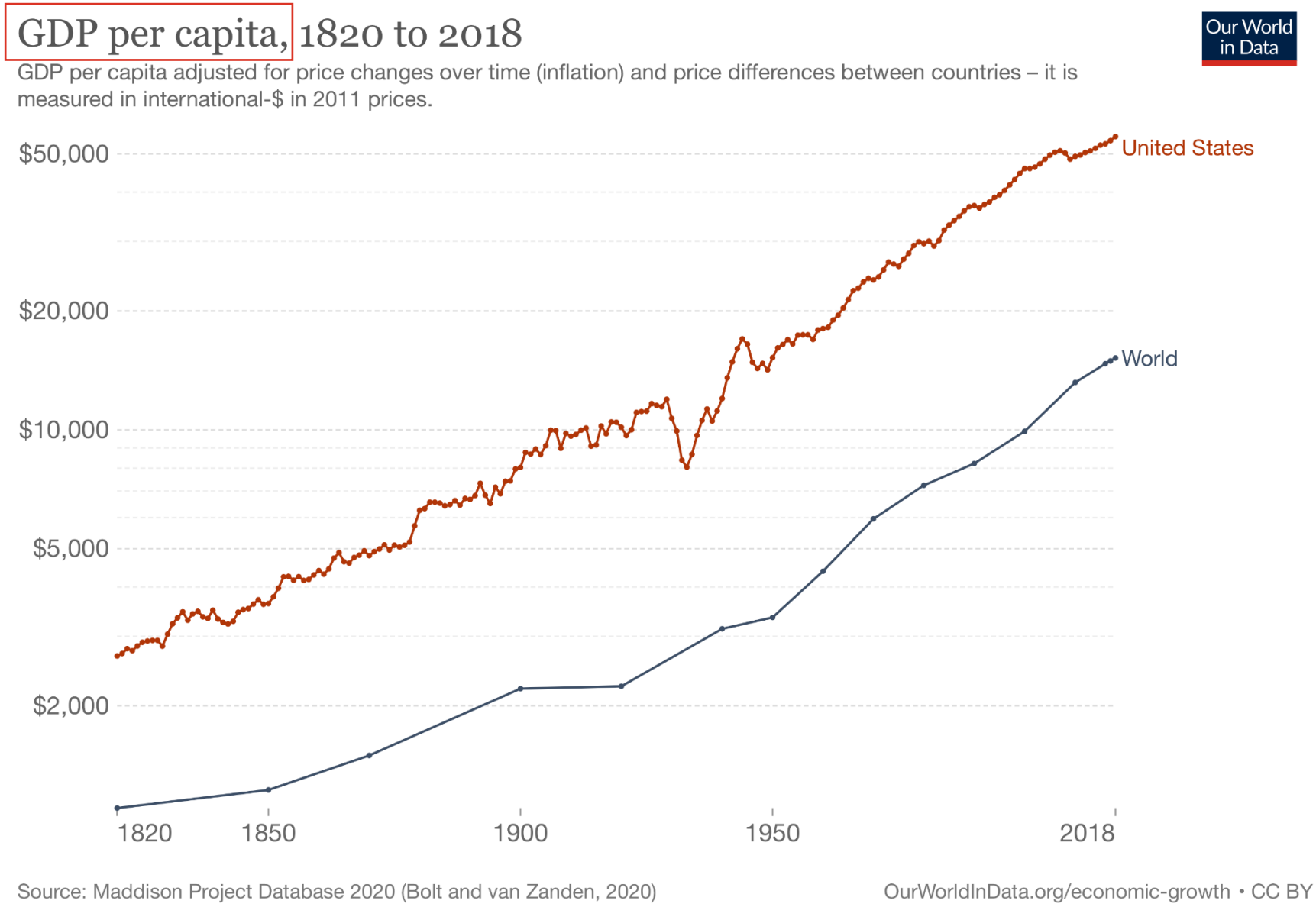
1.
2.
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10.

1a



Our World in Data

1b



Our World in Data

2a

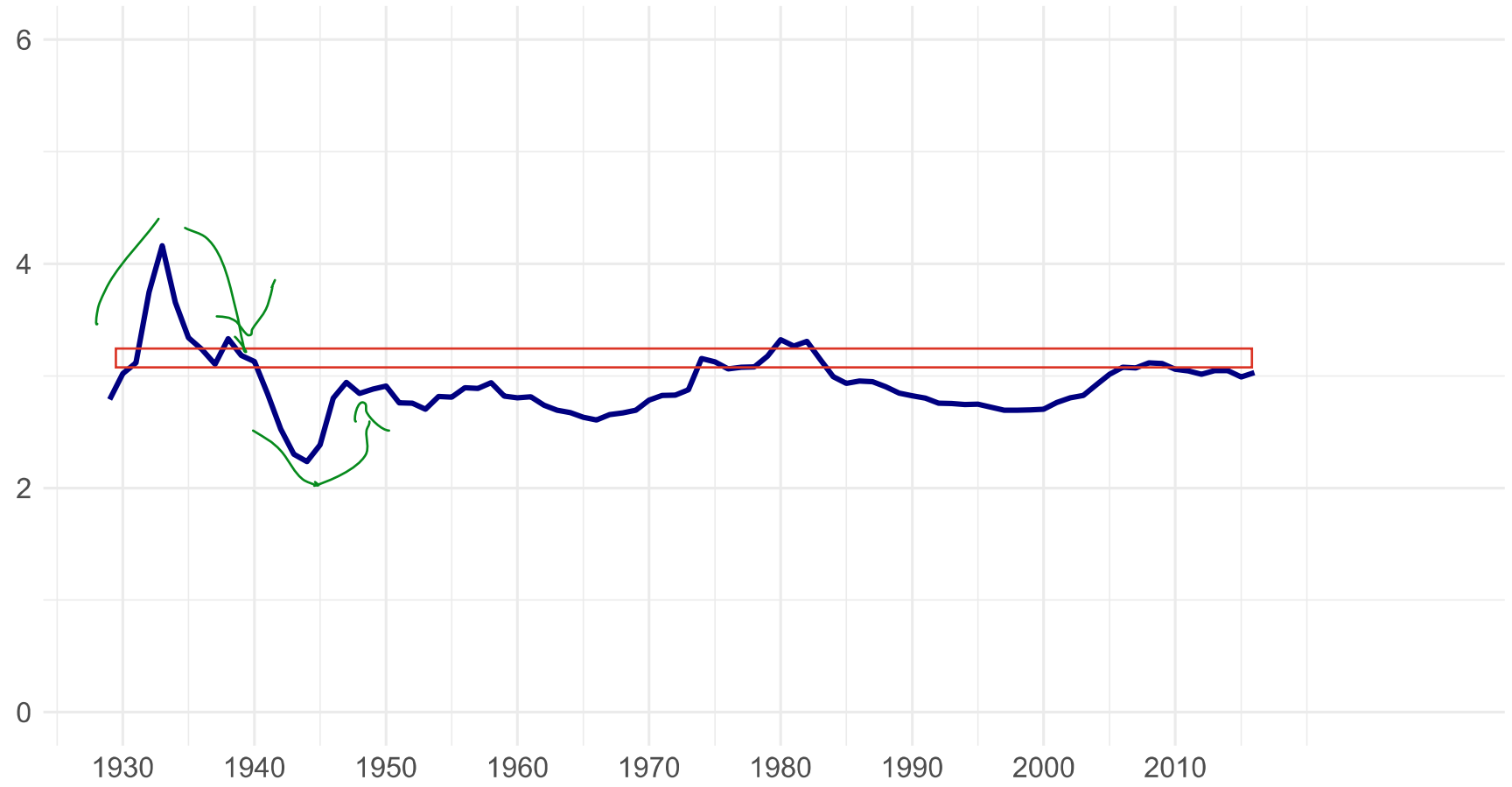


FIGURE 2

Capital–output ratio in the U.S. Ratio of fixed capital and consumer durables to GDP.
Source: NIPA table 1.1.

2b

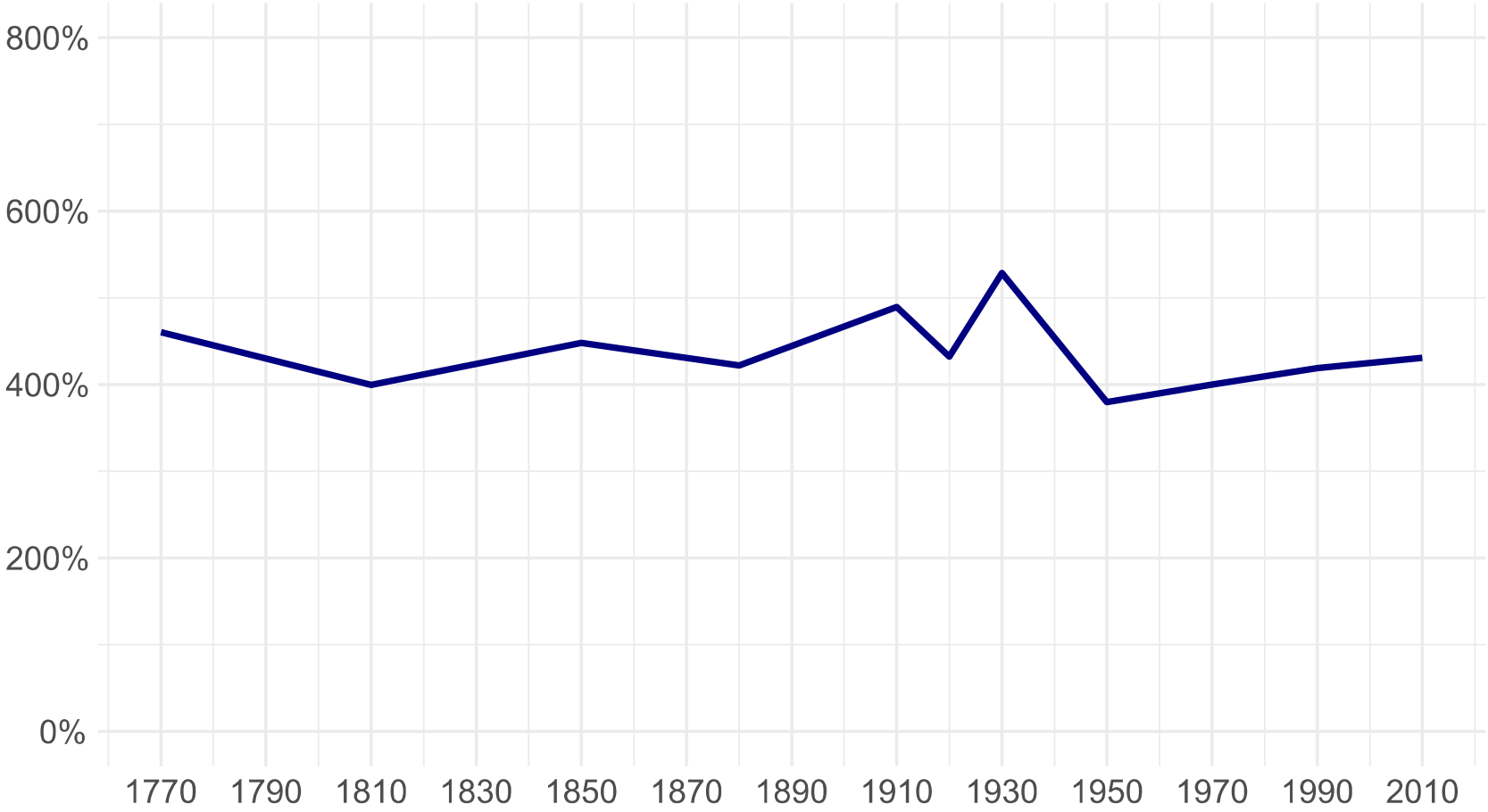


FIGURE 3

Wealth–output ratio in the U.S., historical data. Source: Piketty (2013).

3.

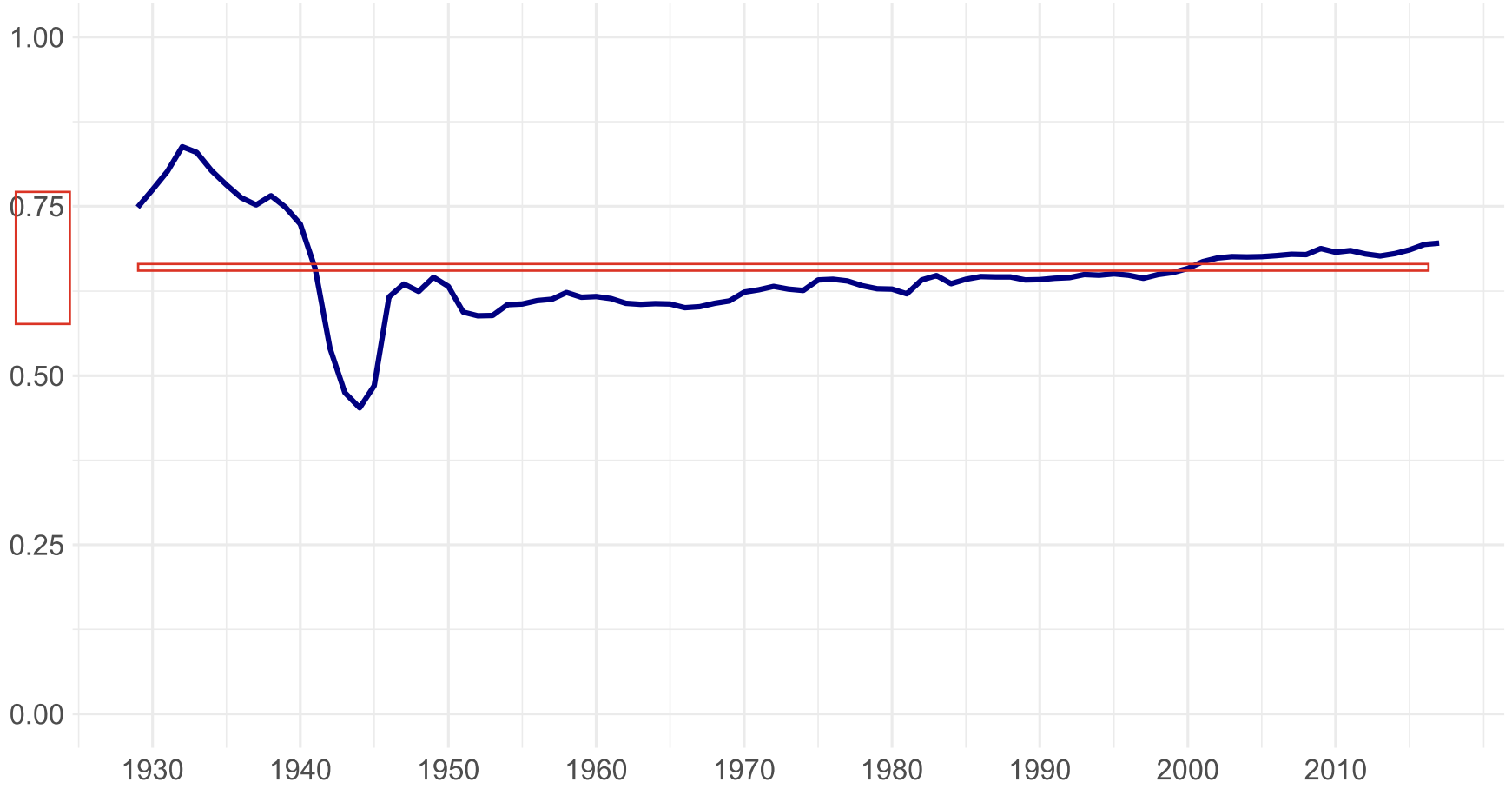


FIGURE 4

Consumption–output ratio in the U.S. The ratio of private consumption to GDP 1929–2017. Source: NIPA table 1.1.5.

4a

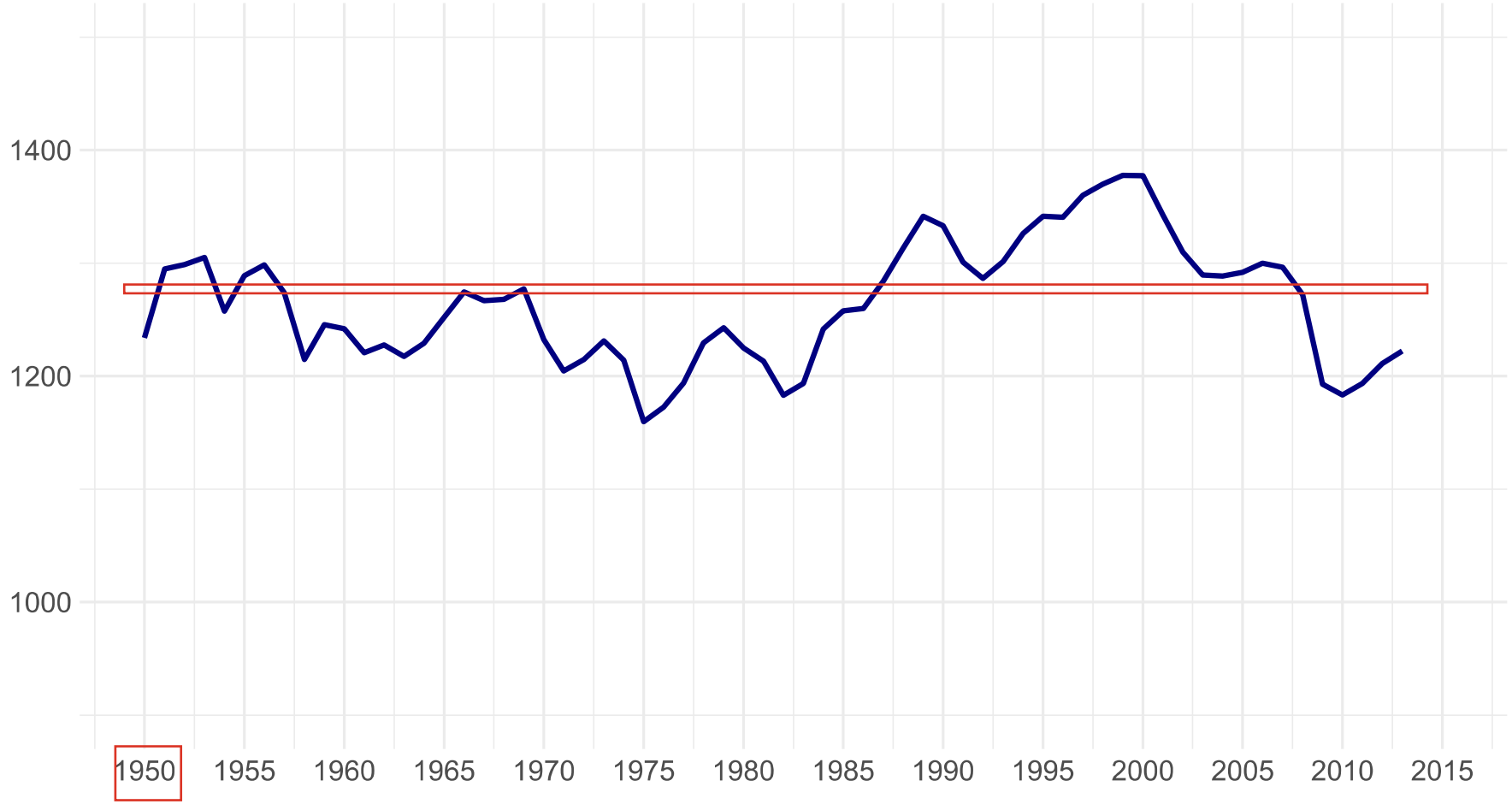


FIGURE 5

U.S. hours worked, postwar period. Average annual hours worked. Source: Boppart and Krusell (2018).

4b

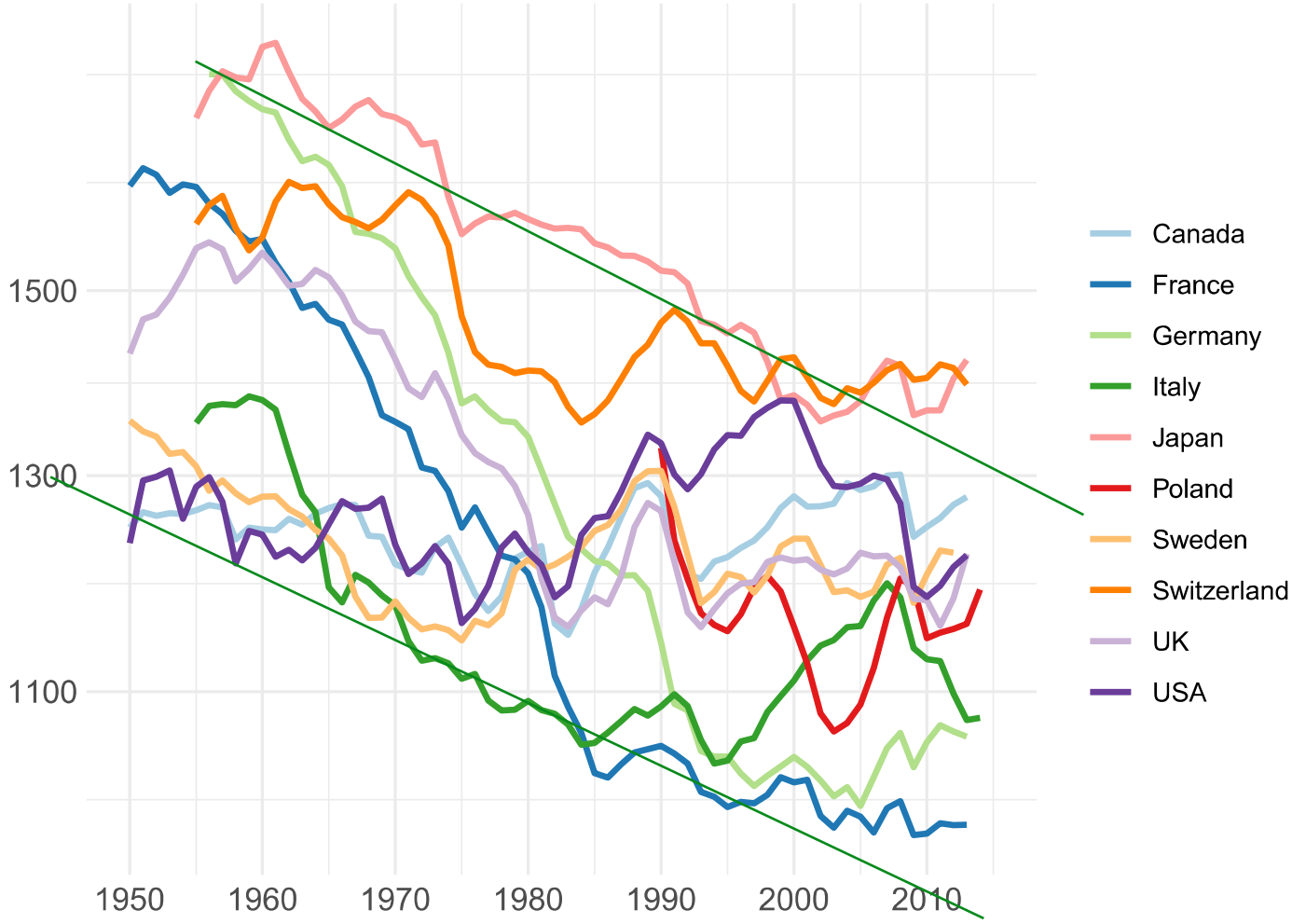


FIGURE 6

Hours worked in the OECD, postwar period. Average annual hours worked. Source: Boppart and Krusell (2018).

5a

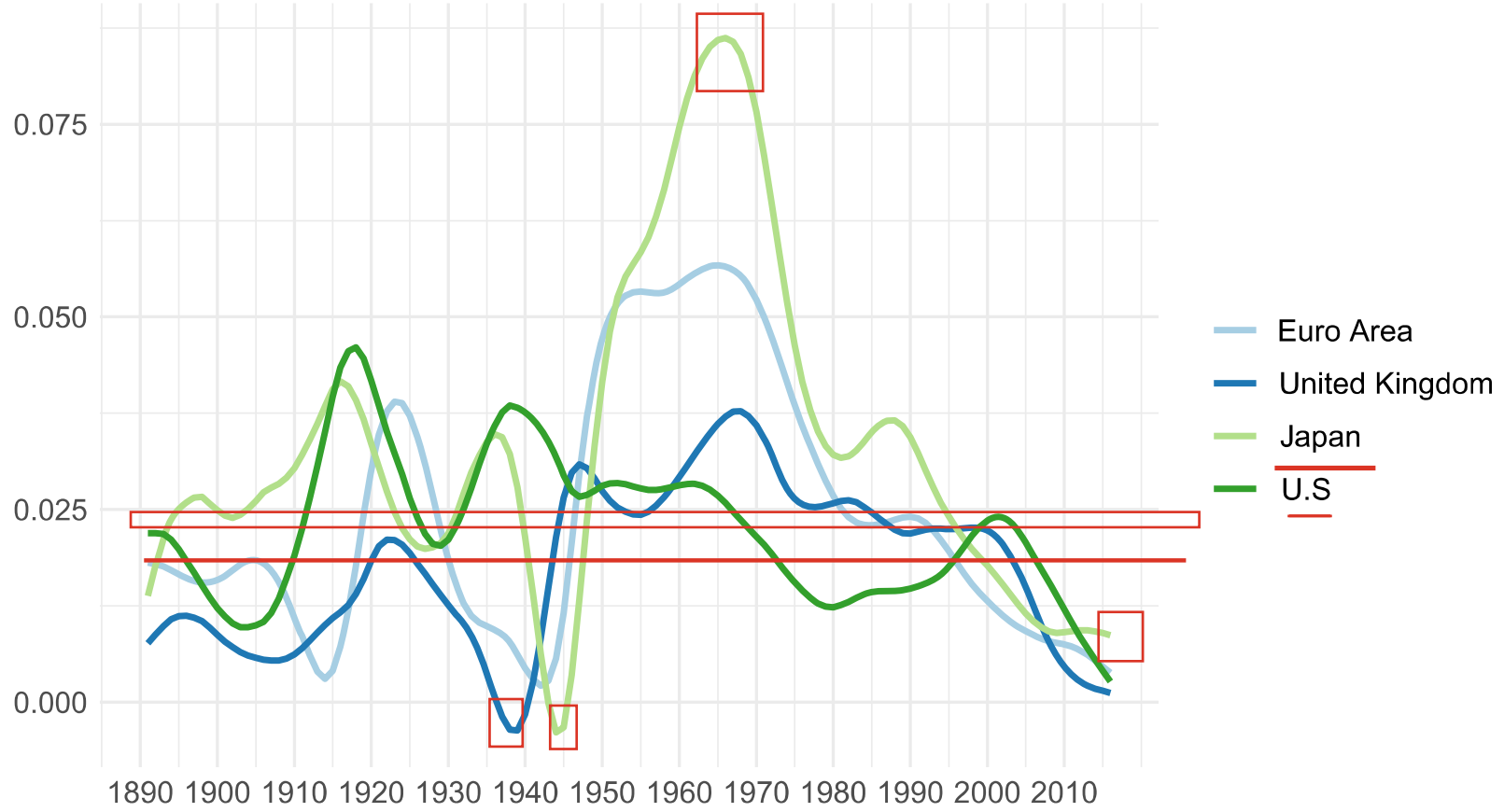
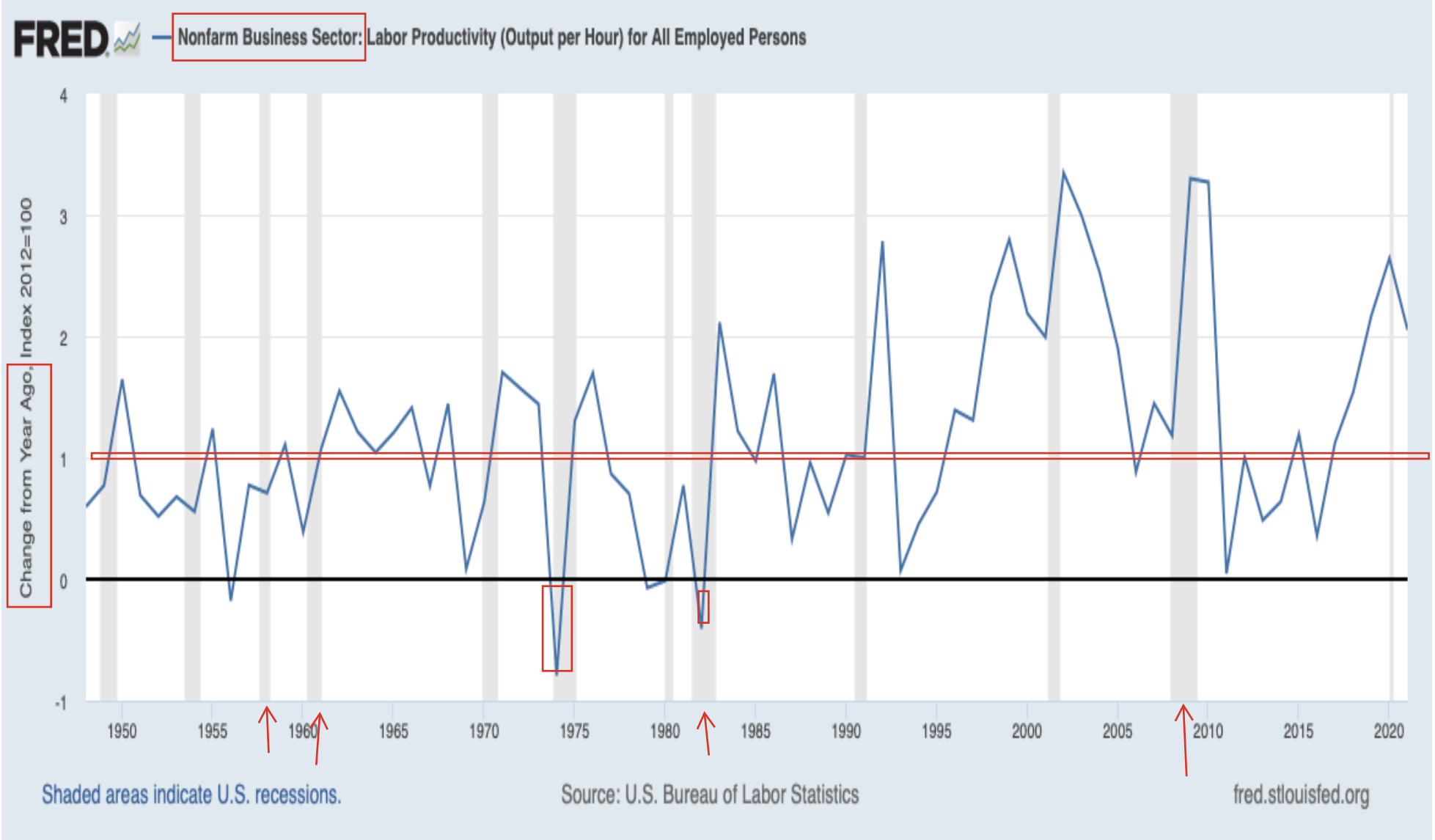


FIGURE 8

Labor productivity for a selection of countries. Hodrick–Prescott-filtered annual growth of labor productivity per hour U.S., the Euro Area, Japan, and the United Kingdom, 1891–2012. Source: Bergeaud et al. (2016).

5b


FRED
Nonfarm Business Sector: Labor Productivity (Output per Hour) for All Employed Persons
Change from Year Ago, Index 2012=100
4
3
2
1
0
-1
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
Shaded areas indicate U.S. recessions.
Source: U.S. Bureau of Labor Statistics
fred.stlouisfed.org

FRED

6a

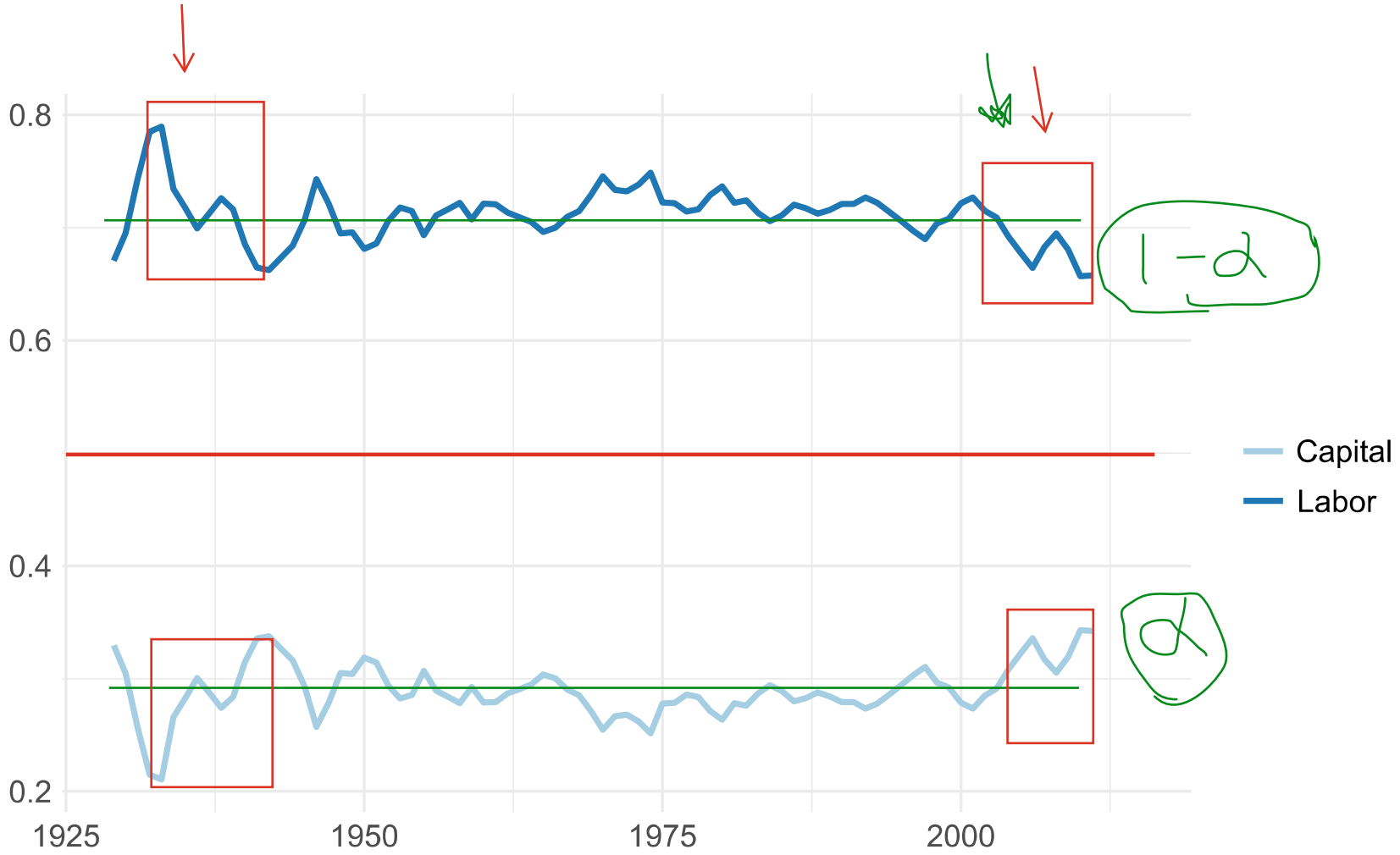


FIGURE 13

U.S. factor shares. Source: updated version of Piketty and Saez (2006).

6b

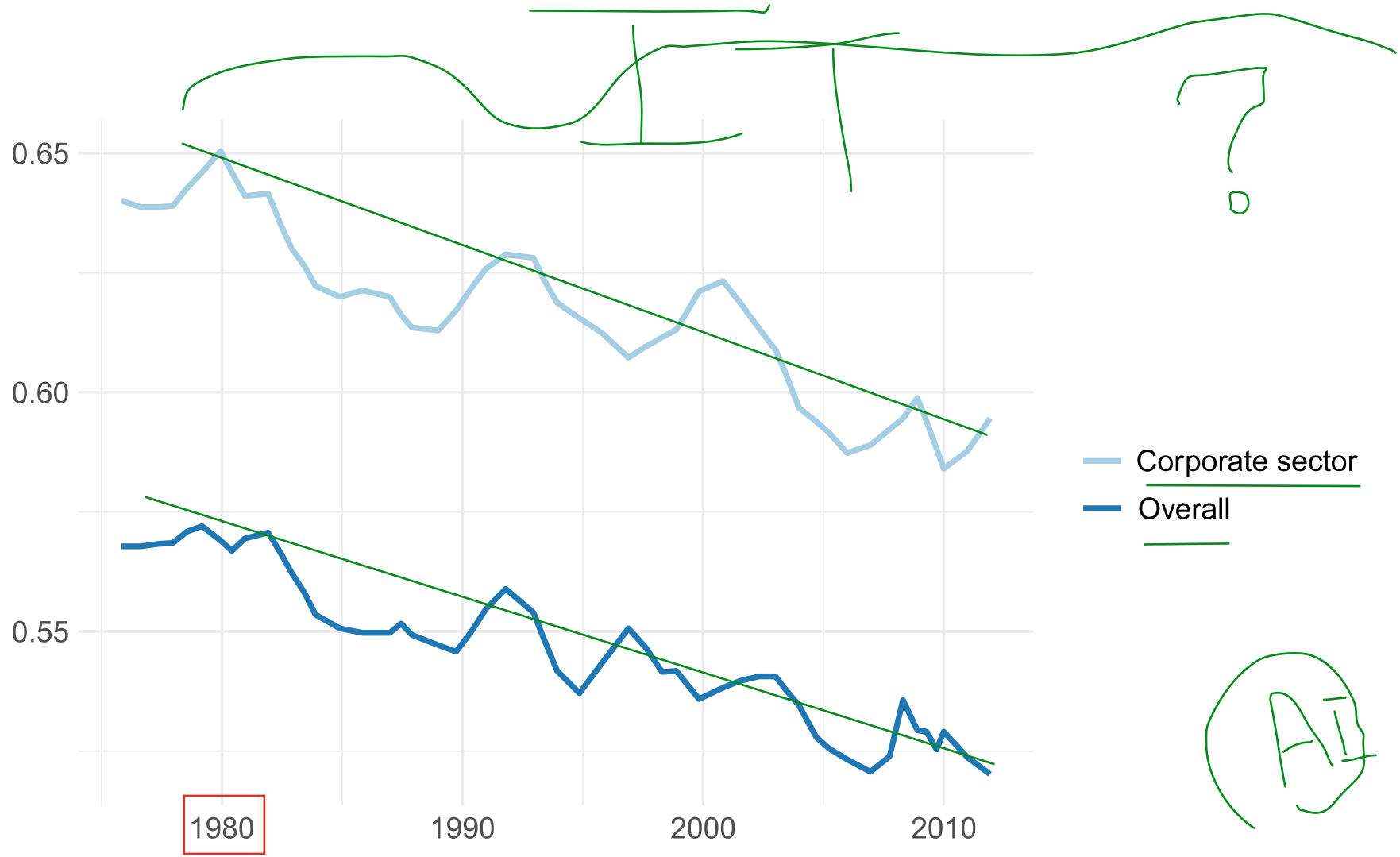


FIGURE 15

The global labor share. Source: Karabarbounis and Neiman (2014).

7a

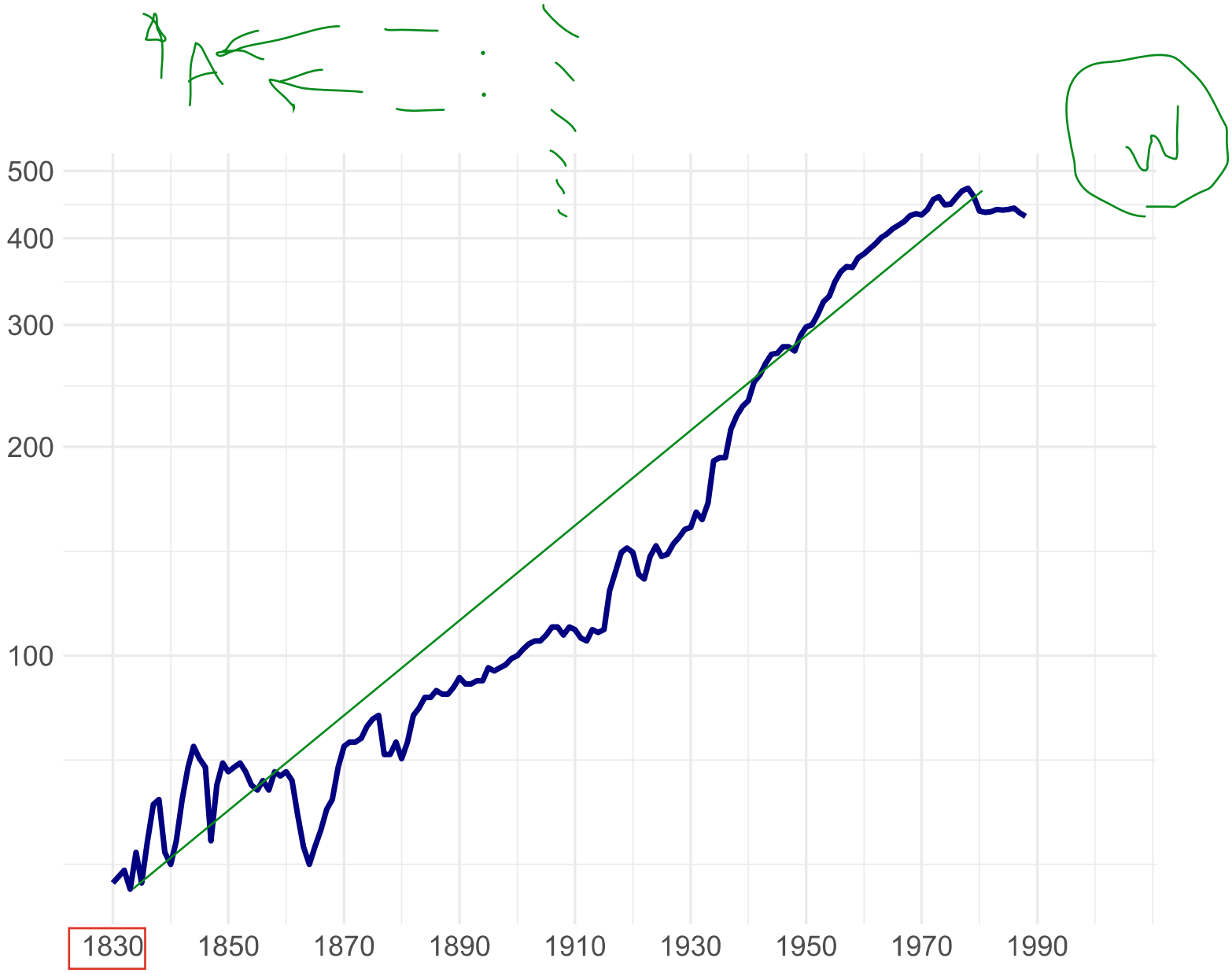


FIGURE 12

Historical real wages in the U.S. Source: Williamson (1995, Table A1.1).

7b

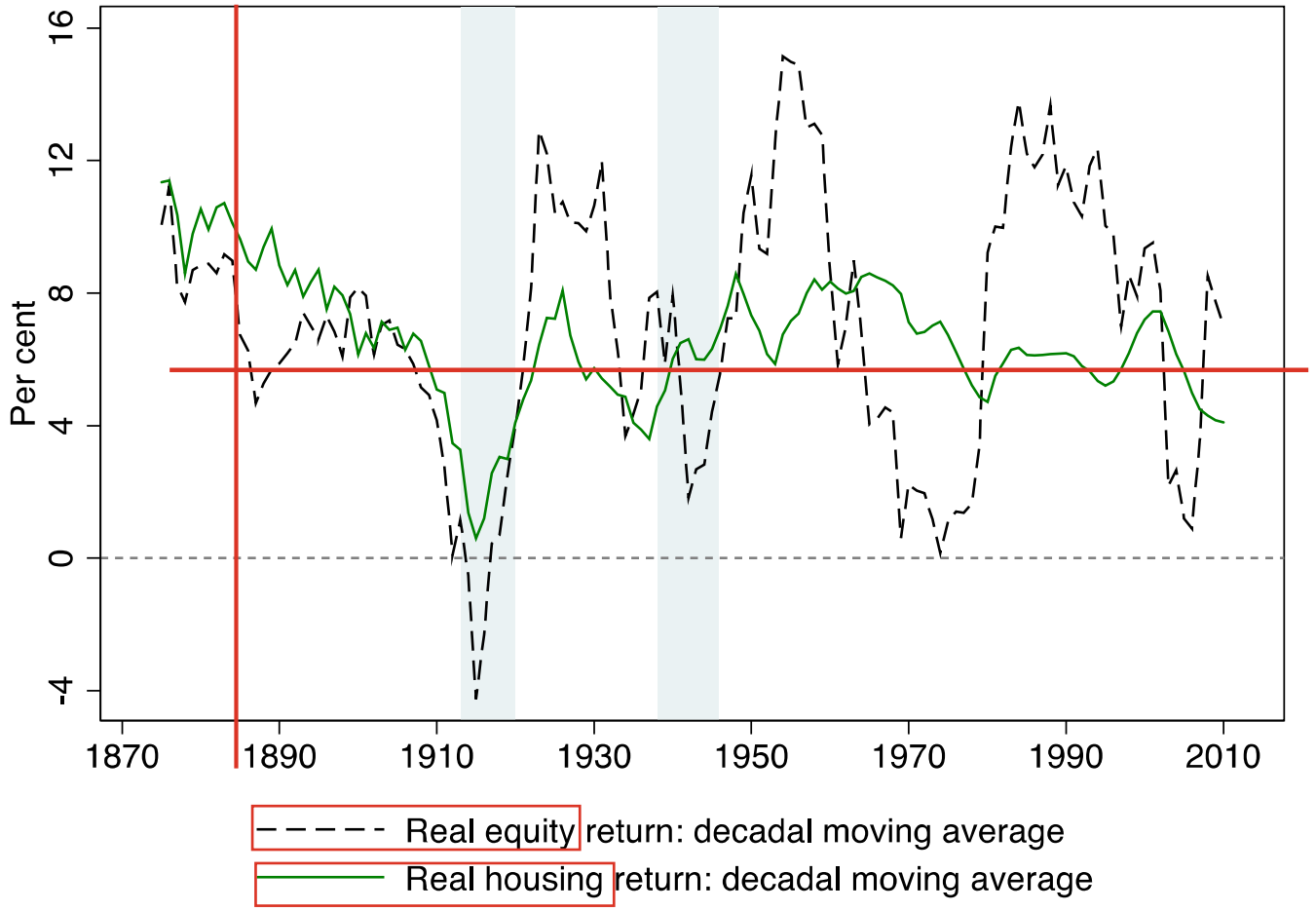


FIGURE VII

Trends in Real Returns on Equity and Housing

Mean returns for 16 countries, weighted by real GDP. Decadal moving averages.

Jorda et al (2019)

8a.

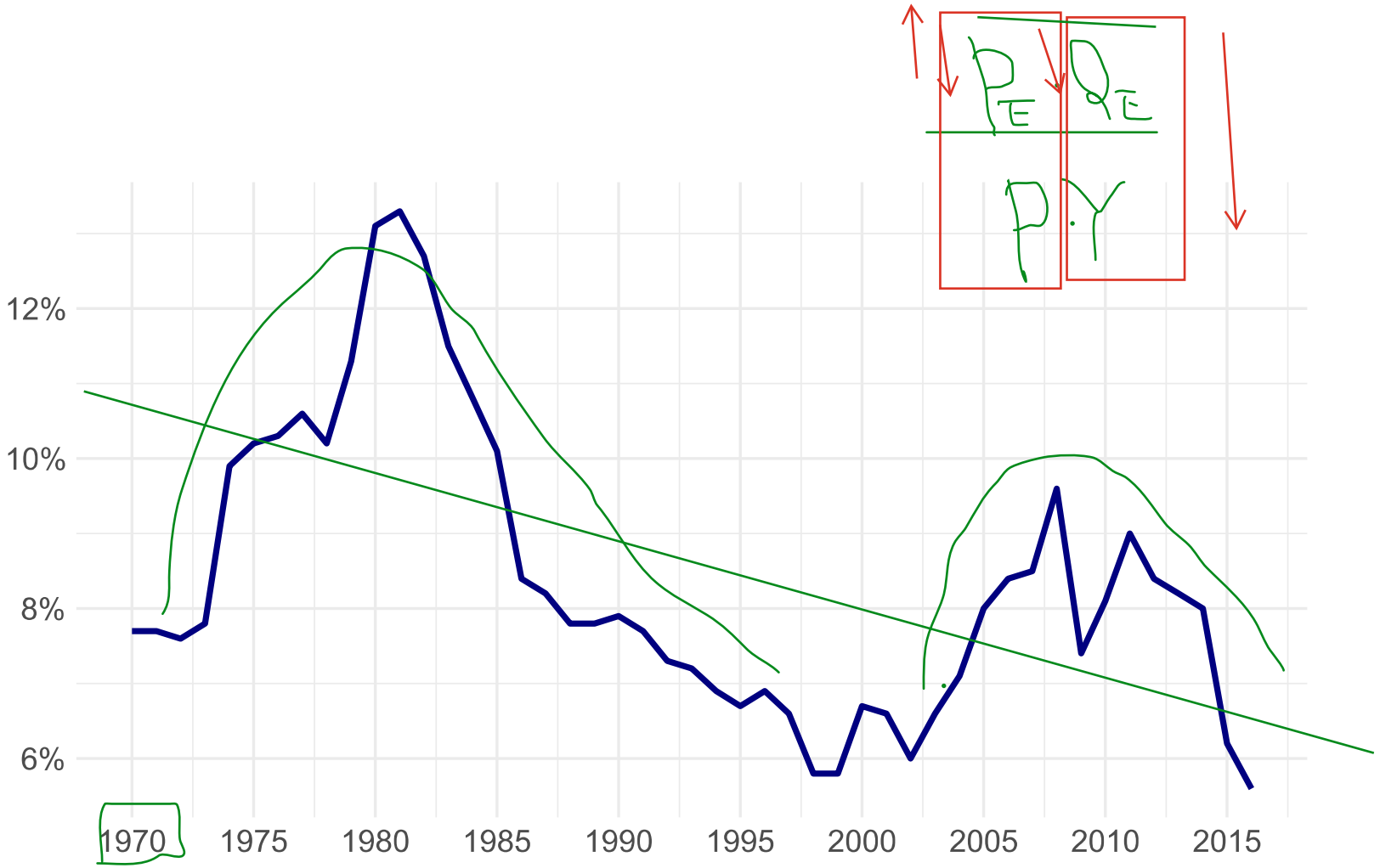
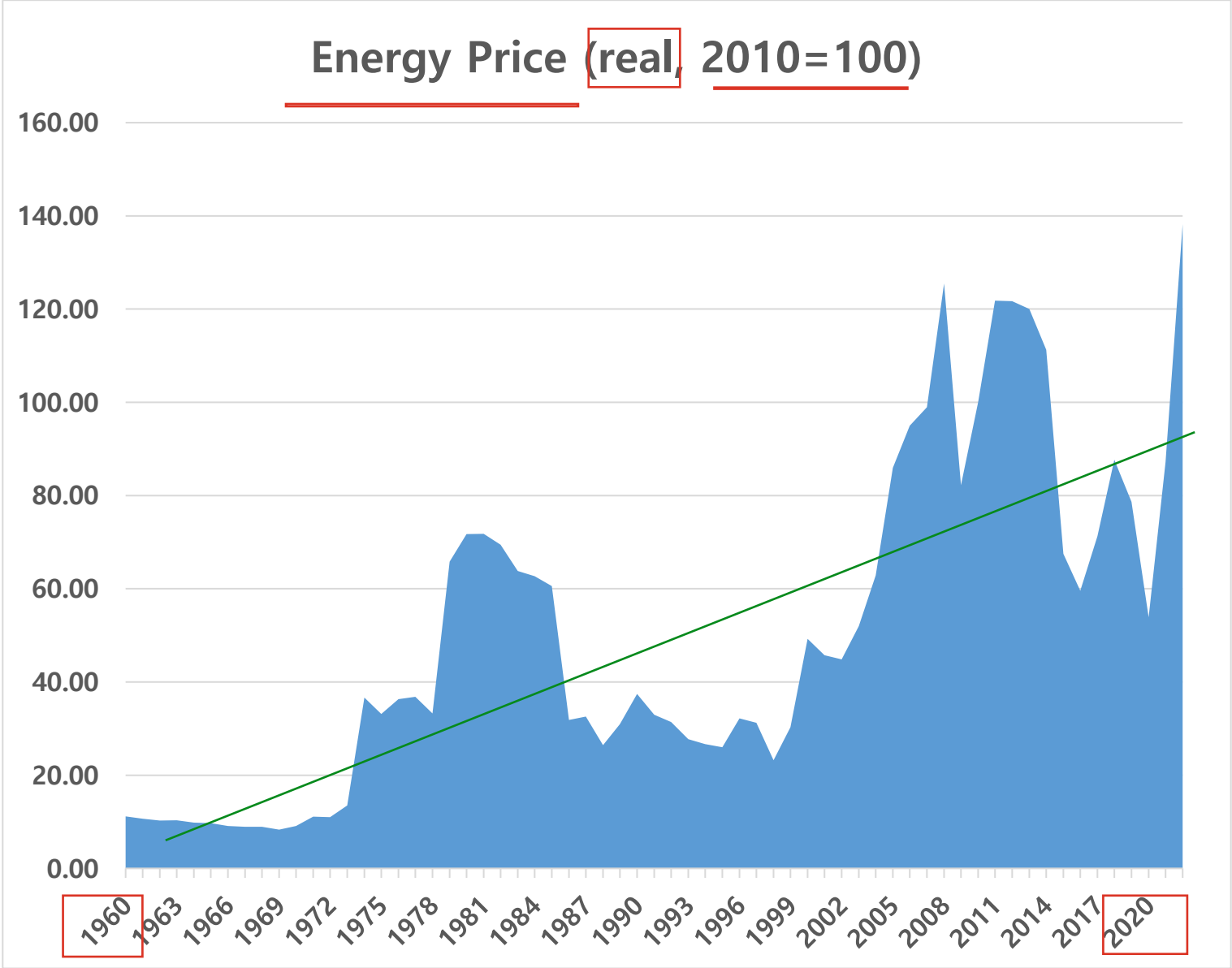


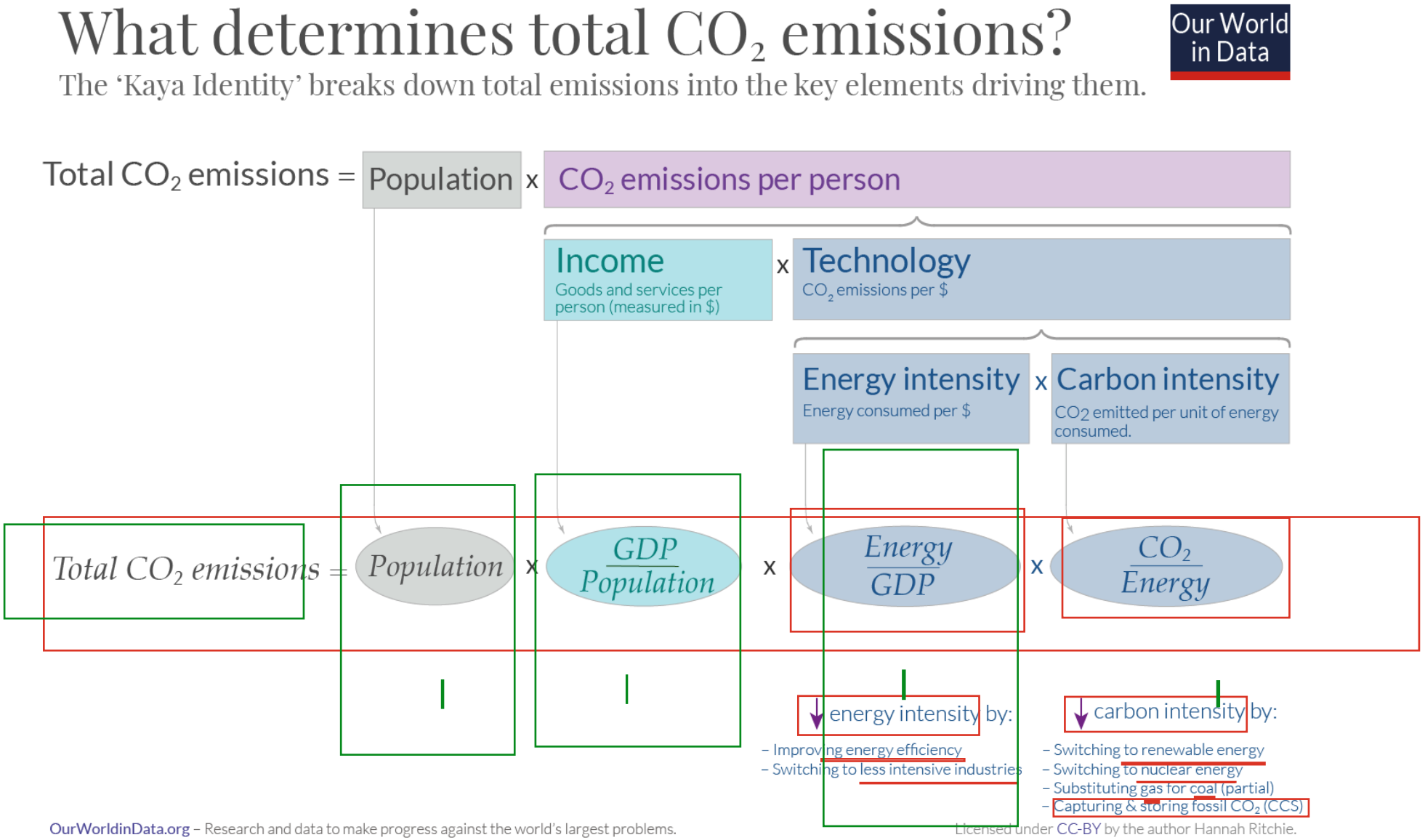
FIGURE 20

The energy share in the U.S. The total nominal energy bill divided by nominal GDP. Source: US Energy Information Administration.

8b.

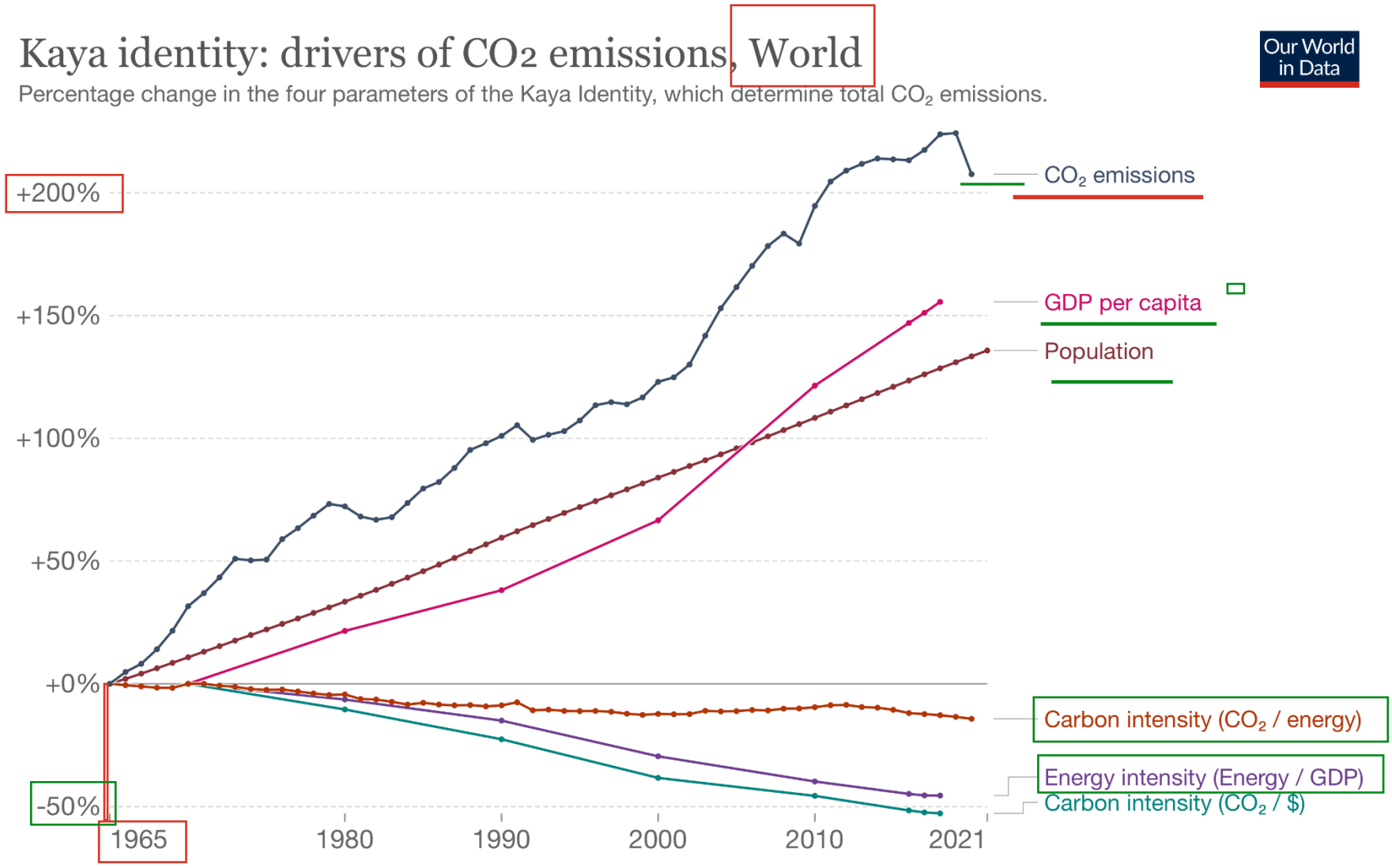

Energy Price (real, 2010=100)
160.00
140.00
120.00
100.00
80.00
60.00
40.00
20.00
0.00
1960
1963
1966
1969
1972
1975
1978
1981
1984
1987
1990
1993
1996
1999
2002
2005
2008
2011
2014
2017
2020

9. Kaya Identity



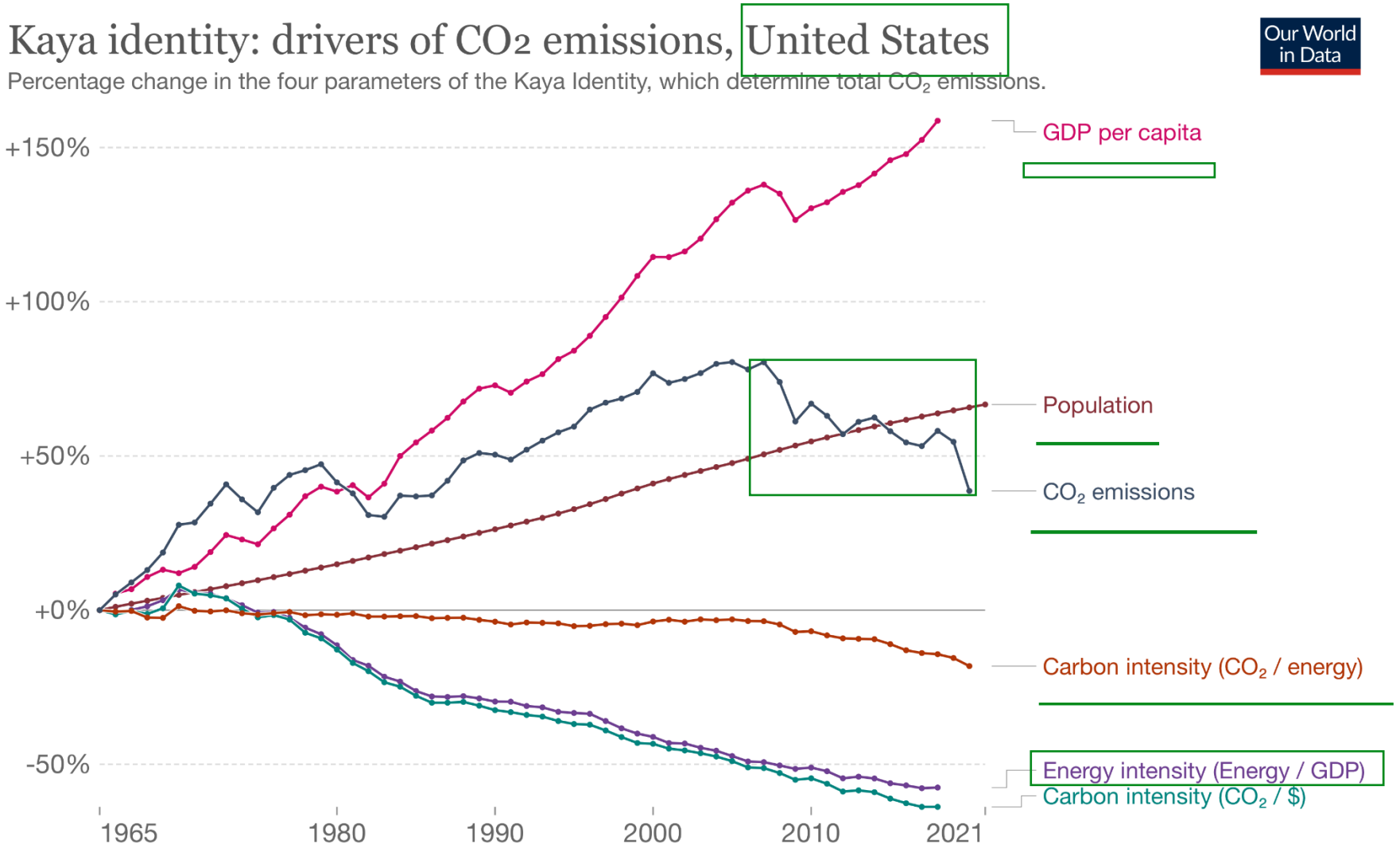
Our World in Data

9a



Our World in Data

9b



Source: Our World in Data based on Global Carbon Project; UN; BP; World Bank; Maddison Project Database
Note: GDP per capita is measured in 2011 international-$ (PPP). This adjusts for inflation and cross-country price differences.
OurWorldInData.org/co2-and-other-greenhouse-gas-emissions • CC BY

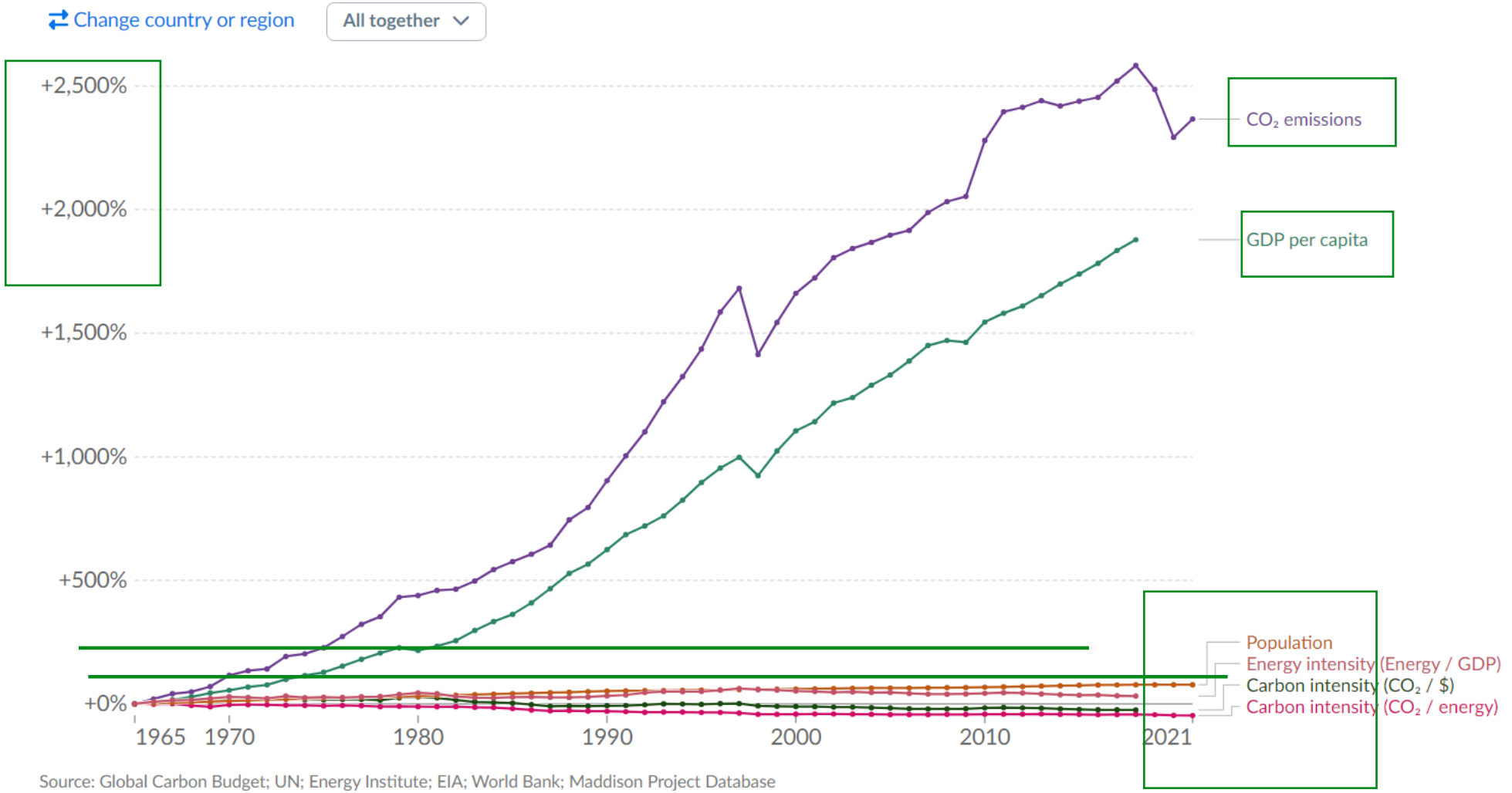
Our World in Data

9c

Kaya identity: drivers of CO_2 emissions, South Korea

Percentage change in the four parameters of the Kaya Identity, which determine total CO_2 emissions. Emissions include fossil fuel and industry emissions. Land use change is not included.

Our World in Data



Source: Global Carbon Budget; UN; Energy Institute; EIA; World Bank; Maddison Project Database
Note: GDP per capita is measured in 2011 international-$ (PPP). This adjusts for inflation and cross-country price differences.
OurWorldInData.org/co2-and-greenhouse-gas-emissions • CC BY

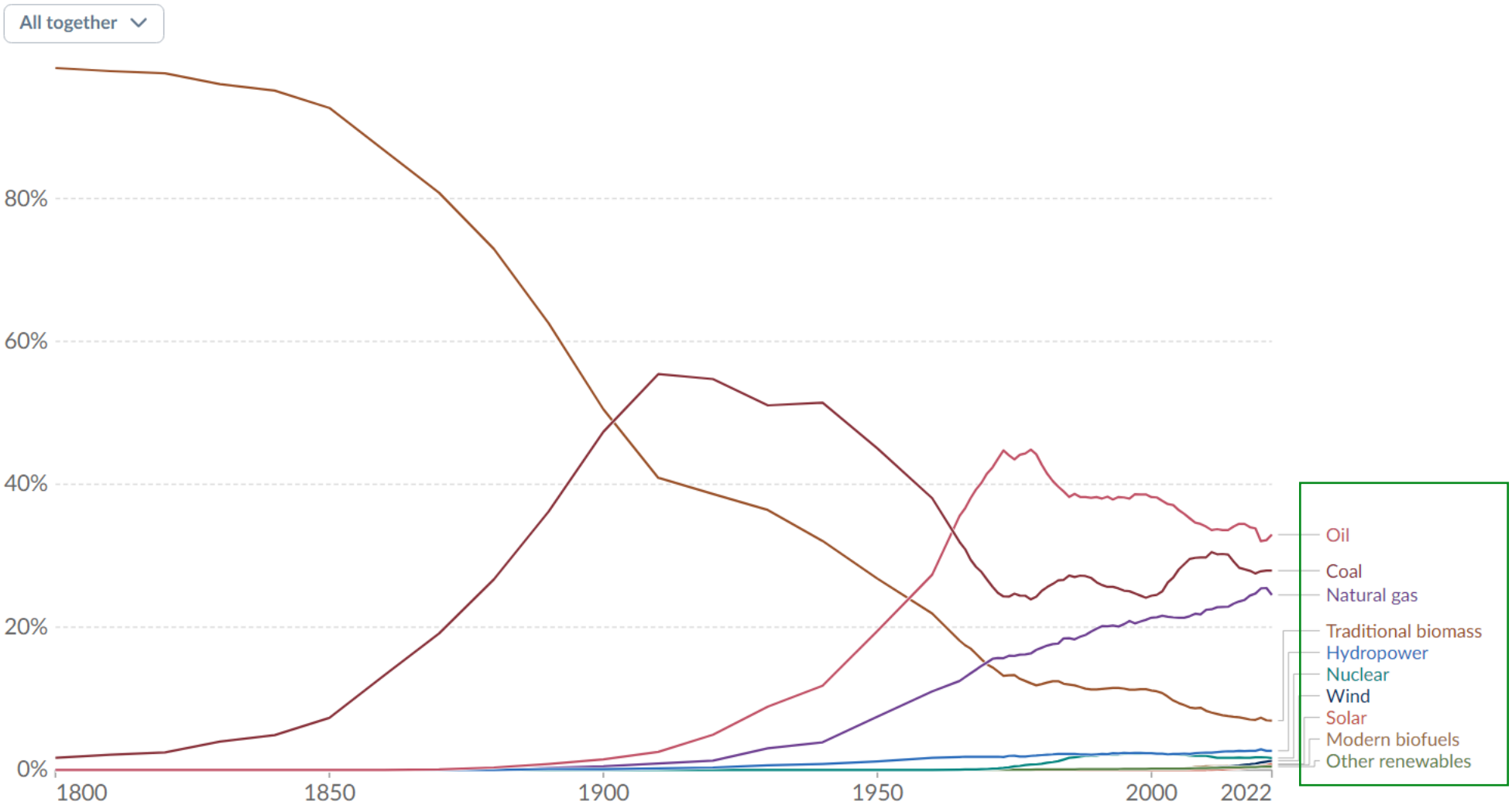
Our World in Data

10a

Share of global primary energy consumption by source

Global primary energy consumption, disaggregated by source. Traditional biomass as an energy source is included here. 'Other renewables' represents all renewable sources minus solar, wind, hydropower, traditional biomass and modern biofuels (e.g. geothermal, wave and tidal).

Our World in Data

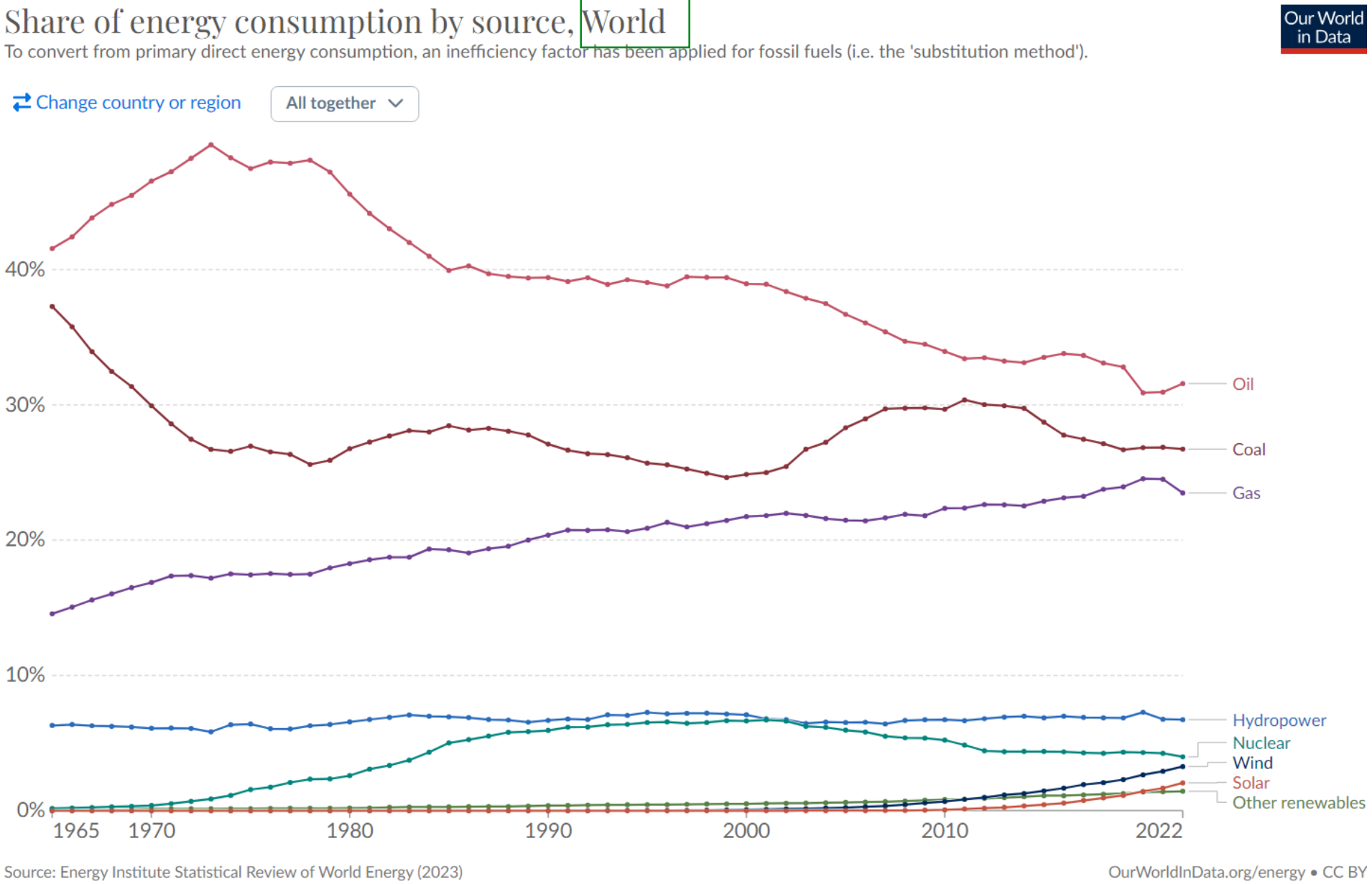


Source: Energy Institute Statistical Review of World Energy (2023); Vaclav Smil (2017)

OurWorldInData.org/energy • CC BY

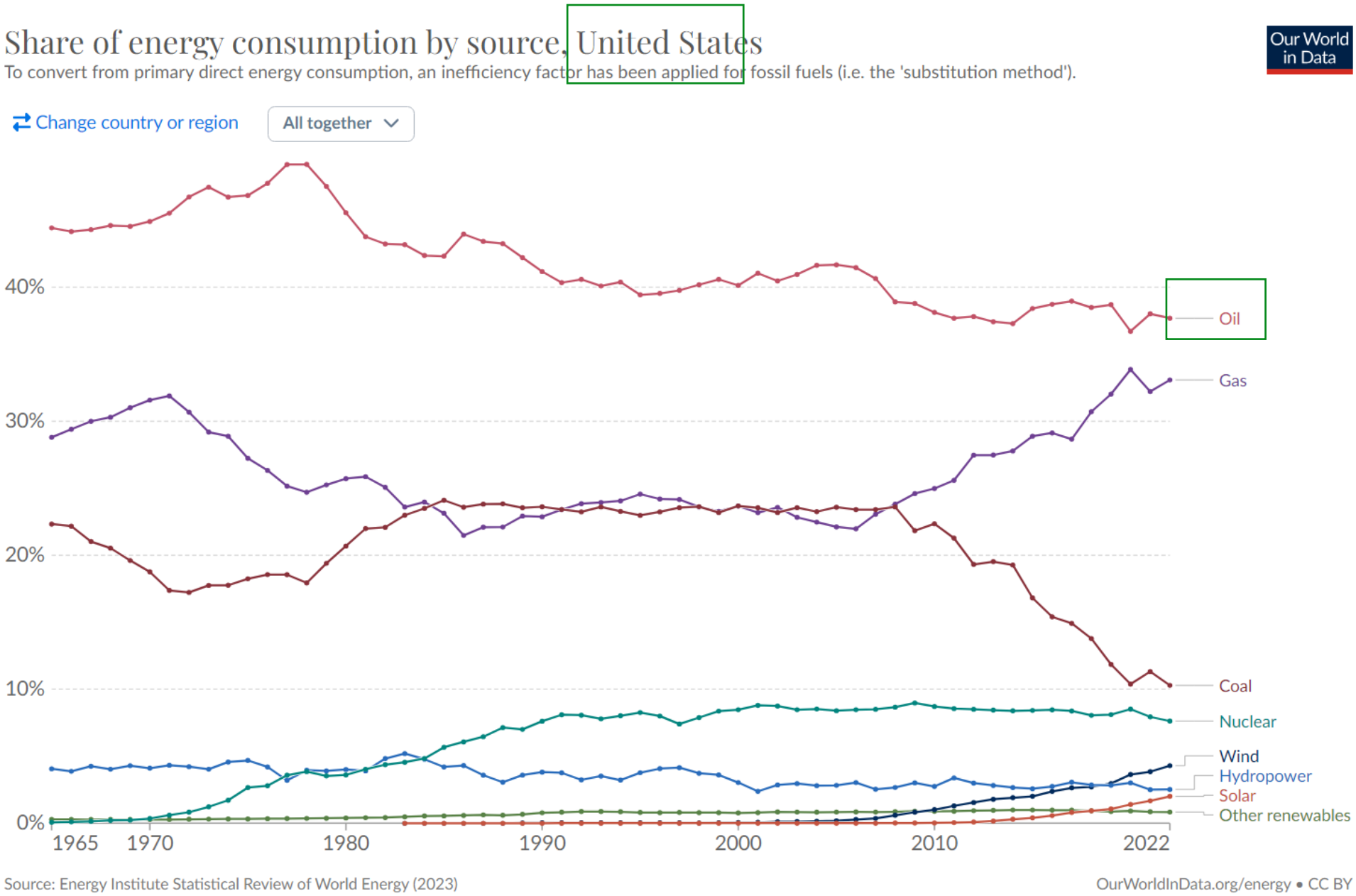
Our World in Data

10a`



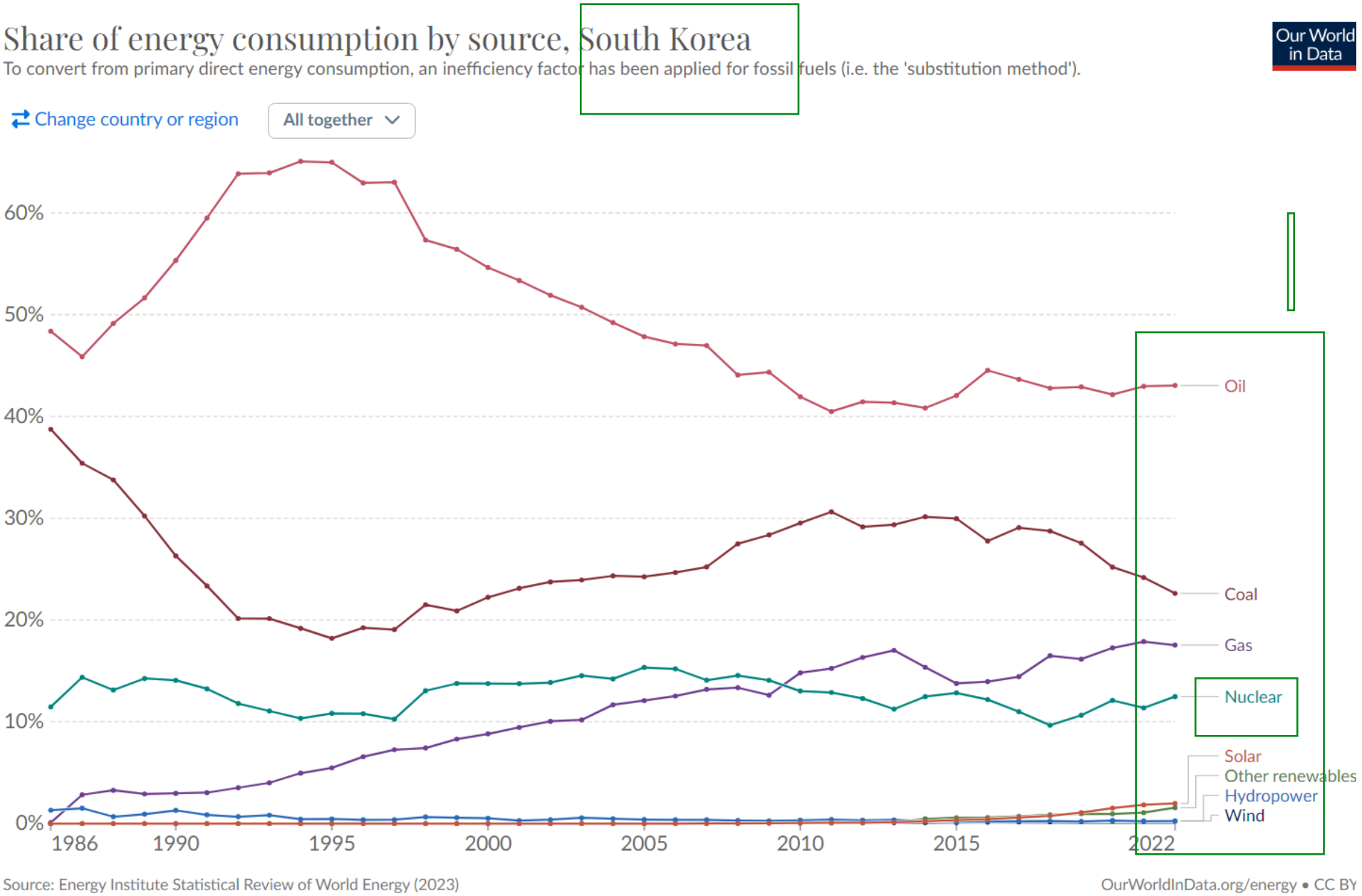
Our World in Data

10b



Our World in Data

10c



Our World in Data